

DESIGNING A ROBUST ALGORITHMIC TRADING STRATEGY

This document outlines my algorithmic trading strategy for the FEC Quant Problem Statement. The backtester starts with \$1,000,000 and uses historical price data to make automated, rule-based trading decisions. Below, I explain my specific parameter choices, prove the model is mathematically robust against overfitting, and share the final performance metrics

PHASE 1:

The initial phase of the strategy focuses on securing the high quality data and continuous historical dataset extraction using a rolling 7 year window for target using appropriate tickers as mentioned in the task using yfinance library.

To maintain the integrity of the performance metrics, the algorithm worked on "Adjusted close" price, filtering the noise.

Furthermore, the dataset was listed and checked for null values using `isnull().mean() * 100` (percentage null values) and accordingly filled using `.ffill().bfill()` to ensure seamlessness ahead. `Adj_close_df` and `volume_df` were extracted accordingly and listed using `.head()` & `.info()` for checking for appropriate dtypes and entries and the phase 1 was as such completed

PHASE 2:

One of the crucial phase where a certain set of rules ie signals were made so as to tell the algorithm whether a certain stock is worth buying, holding or to be sold.

Here, the technical indicators that I used were :

[1] 50 Day & 200 Day SMA (ans to 'Is the stock fundamentally going up?')

Raw daily stock prices are highly erratic and full of noise and thus comparison based on simply moving average ie SMA against a slow moving 200 day filters out the noise and identifies the true structural market shifts

50 SMA > 200 SMA :

This signals that recent price action is overwhelmingly stronger than the historical average, indicating a confirmed, long-term macroeconomic uptrend.

50 SMA < 200 SMA:

This signals a macro collapse, effectively acting as a circuit breaker to halt long positions.

[2] 14 DAY RSI (Momentum confirmation n ans to 'Is it safe to enter right now or is this a fake rally?')

The Relative Strength Index (RSI) measures the velocity and magnitude of recent price changes.

RSI > 50: demands proof that bullish momentum is actively driving the stock higher at the exact moment of the crossover

[3] 20 DAY ROLLING VWAP(Ans to 'Are institutional investors actually participating in this trend?')

Standard moving averages only look at the closing price. The Volume Weighted Average Price (VWAP) factors in the actual trading volume, making it the ultimate proxy for institutional liquidity. VWAP acts as the Dynamic Exit/Profit-Taking Trigger ie

If it follows the [1] but the price drops below the 20 day VWAP, it means that the heavy buyers have stopped defending the price, thus the algo interprets this as structural support and moves the position to cash(0) to protect the profit.

[4] 14 DAY ATR(Ans to 'How wide should our safety nets be based on current market panic?')

The Average True Range (ATR) is a pure volatility indicator; it does not care about trend direction, only about how violently the asset is swinging.

By calculating the 14-day ATR, the algorithm establishes dynamic, breathing risk parameters. It widens the stop-loss during highly volatile periods to avoid getting shaken out by random noise, and tightens the stops during quiet markets to strictly protect capital.

[PLEASE NOTE : okay so when i implemented sma over 200 days; the thing was for starting days the table filled up with zeroes as here may 30, 2019 the first row of dataset was taken upon and looked backwards but theres no data present right so first 199 rows in 2019 were kinda warm up period and signals fire untill roughly march 2020 so the code below works just fine normally with its first 199 rows in 2019 as warmup period, otherwise to evaluate the active performance n handle this problem, chopping off the periods before calculating the performance; the below code exhibits this]

PHASE 3:

The backtesting was carried out; the results are as follows :

Final Portfolio Value : \$2,276,045.89

Total Sweep Interest : \$211,852.13

CAGR : 14.21%

Sharpe Ratio : 0.72

Max Drawdown : -33.42%

Average Monthly Trnvr : 80.83%

PHASE 4:

To avoid overfitting following was done :

[1] Instead of testing the strategy on the entire 7-year dataset at once, the code slices the timeline into two distinct eras. The first 70% of the timeline is the Train Set (where you would hypothetically optimize your indicators), and the final 30% is the Test Set.

[2] Parameters set to :

SMA_WINDOW = 200

RSI_WINDOW = 14

BUY_RSI = 50

By locking these parameters globally before the function runs, the code forces both the Train and Test backtests to use the exact same mathematical rules.

[3] test_results = generate_strategy_signals(test_close, test_high, test_low, test_volume)

The strategy function is executed entirely independently on the test_variables; The algorithm is forced to trade through the final 30% of the timeline without having seen it during the training phase.

[4] PREVENTING DATA LEAKAGE

execution_signals = daily_signals.shift(1).fillna(0)

shifting the execution signals by one day is critical to preventing the illusion of a perfect model.

[5] Also, train_cagr == test_cagr

Train CAGR : 9.99%

Test CAGR : 10.31%;

thus no overfitting

[6] Also a graph plotted on strategy's equity curve entirely on out of sample data showed that despite the unseen market conditions, there were pullbacks associated demonstrating the growth from starting \$100,000 to \$125,000; the definitive proof that the model is effective against overfitting.

PHASE 5:

This baseline simulates allocating the initial \$1,000,000 evenly across the 10-asset universe and holding it untouched for the entire 7-year period

The evaluation relies on standardised quantitative metrics calculated for both the Train (In-Sample) and Test (Out-of-Sample) datasets:

Annualized Return (CAGR) & Volatility, Sharpe & Calmar Ratios, Maximum Drawdown (MDD), Market Exposure etc worked on and finally, the strategy's stability is visualised through *three graphs*: a *Logarithmic Equity Curve* to track compound growth , a *Drawdown Plot* to visualise the duration and depth of underwater periods , and a Rolling 6-Month Sharpe Ratio to ensure risk-adjusted returns remain consistent across different market regimes rather than relying on isolated lucky trades

BONUS TASK : done in ipynb file